

- 6 (a) A metal plate has work function of 4.5 eV. When ultraviolet radiation of wavelength 120 nm is incident on the plate, a photoelectric current is detected.
- (i) Calculate the maximum speed of the photoelectrons as a result of this illumination.

maximum speed = m s⁻¹ [2]

- (ii) The uncertainty in the speed of the photoelectrons in (a)(i) is 0.50%.

Assuming that the photoelectrons are moving in the horizontal direction,

1. determine the uncertainty in the horizontal momentum of the photoelectrons;

uncertainty = N s [3]

2. determine the minimum uncertainty in the position of the electron.

uncertainty = m [2]

(b) Fig. 6.1 shows an x-ray spectrum.

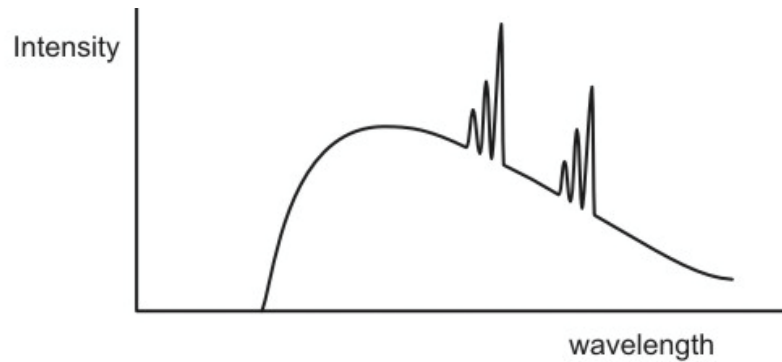


Fig. 6.1

Describe how the continuous spectrum is produced.

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[4]